

Mapping Action Space via Psychophysiological Responses in Virtual Reality

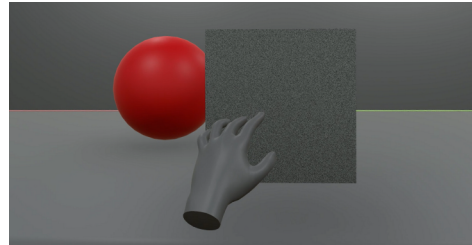
Emma Vicamini^{a,b,*}Antonio Sterlicchio^bAlberto Gallace^{a,b}^aDepartment of Psychology, University of Milan-Bicocca, Milan, Italy^bMind and Behavior Technological Center (MiBTec), Department of Psychology, University of Milan-Bicocca, Milan, Italy

Figure 1: Experimental setup. Participants interact with a central object while task-irrelevant stimuli appear at different distances.

ABSTRACT

‘Action space’ is the region where objects become relevant for human interaction. Using VR, we investigated how the distance of task-irrelevant stimuli modulates behavioral, oculomotor, and physiological responses. Participants performed a dual task involving interaction with a central target while ignoring peripheral distractors presented at different distances. Near and intermediate distractors induced stronger behavioral interference and oculomotor responses than far distractors, revealing a graded spatial organization of action space. Increasing distractor salience through motion produced faster saccades and longer fixations. Distance judgments showed that distractors close to the body were perceived as closer than their actual distance.

Index Terms: Action space, distance perception, oculomotor dynamics

1 INTRODUCTION

Peripersonal space (PPS) refers to the space immediately surrounding the body, where objects are most relevant for interaction. It is supported by well-established neuropsychological mechanisms [2]. PPS is divided into defensive and goal-directed subspaces [1]. While the defensive component has been extensively studied, the goal-directed PPS (here referred to as ‘action space’) remains less understood. Here, we use Virtual Reality (VR) to characterize action space by measuring behavioral, oculomotor, and physiological responses to distractors at three distances: Near (50 cm), Mid (100 cm), and Far (150 cm). We also assessed how these distances are perceived both explicitly and implicitly. This framework provides a baseline for Human Augmentation research, exploring how action space adapts when motor reach is digitally extended (e.g., via virtual supernumerary or elongated limbs).

2 METHODS

2.1 Participants & Setup

Experiment 1 included 27 participants (16 F, 11 M; mean age = 24.8 ± 4.4 y, range = 20–41). Experiment 2 reports preliminary

data from 13 participants (9 F, 4 M; mean age = 22.9 ± 2.3 y, range = 20–28).

Data were collected using a Varjo VR-3 HMD with integrated eye tracking (200 Hz). We analyzed behavioral (reaction times, accuracy), oculomotor (saccadic latency, fixation duration, probability of distractor fixation), and physiological measures (pupil diameter variation, blink rate).

2.2 Task & Design

Participants performed a dual-task paradigm, touching a visual anomaly on a central cube (primary task) while detecting a lateral distractor (secondary task) presented at three distances (50, 100, 150 cm) (Fig. 1).

Experiment 1 employed a $2 \times 2 \times 3$ within-subject design (336 trials), with Anomaly Presence (Yes/No), Distractor Presence (Yes/No), and Distractor Distance (Near, Mid, Far). Trials without an Anomaly served as a baseline to assess distractor processing without primary task interference.

Experiment 2 (608 trials) introduced an additional Distractor Motion factor (Static, Lateral Motion at $10\text{--}12^\circ/\text{s}$, Jittering). Motion parameters were selected to engage the magnocellular dorsal stream [3].

At the end of both experiments, distance perception was measured via explicit verbal reports and implicit motor estimates; in the latter, participants positioned themselves relative to a wall to match the distractor’s perceived location.

3 KEY FINDINGS

3.1 Experiment 1

- RTs for anomaly detection were modulated by distractor distance, $\chi^2(3) = 19.70$, $p < .001$: Near and Mid distractors slowed responses relative to the no-distractor baseline (Near vs. None: $p < .001$; Mid vs. None: $p = .002$), whereas Far did not (Far vs. None: $p = .198$). Interference decreased linearly with distance ($p = .014$).
- The probability of fixating the distractor was rare ($\approx 1.5\text{--}6.5\%$) but strongly distance-dependent, $\chi^2(2) = 39.60$, $p < .001$. Fixation probability was higher for Near and Mid than Far (both $p < .001$), with no Near–Mid difference ($p = .166$), following a linear spatial gradient ($p < .001$), independent of anomaly presence (Fig. 2).

*Contact: e.vicamini1@campus.unimib.it

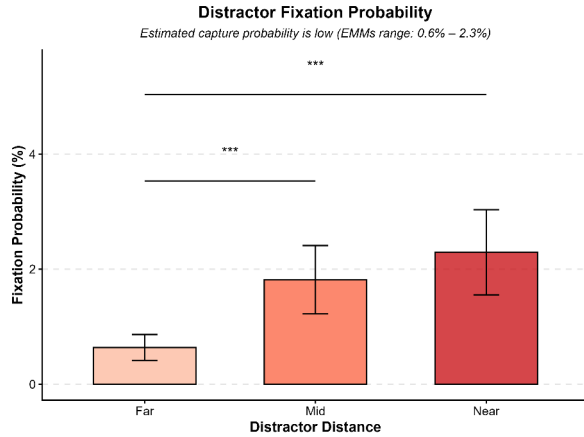


Figure 2: Exp.1 - Probability of fixating the distractor as a function of distance. Near and Mid distances elicit higher fixation probability than Far distance.

- Saccadic latency was modulated by both anomaly presence, $\chi^2(1) = 29.09$, $p < .001$, and distractor distance, $\chi^2(2) = 6.24$, $p = .044$. Saccades were faster toward Near than Mid distractors (Near vs. Mid: $p = .022$), with no linear spatial trend ($p = .250$). Fixation duration was reduced by anomaly presence, $\chi^2(1) = 29.26$, $p < .001$, regardless of distance.
- Distance perception was distorted, $\chi^2(2) = 29.17$, $p < .001$, with no effect of response modality ($p = .121$) and no interaction ($p = .803$). Near space was underestimated (-34.2% , $p = .026$), Far space overestimated ($+33.3\%$, $p = .030$), while Mid distance was accurate ($p = .738$) (Fig. 3).

3.2 Experiment 2

- Distractor fixation probability replicated Experiment 1, showing a significant main effect of distractor distance, $\chi^2(2) = 40.20$, $p < .001$. Fixation probability was higher for Near and Mid compared to Far distractors (both $p < .001$), with no difference between Near and Mid ($p = .986$), and independent of distractor motion and anomaly presence.
- Saccadic latency showed a Distractor Distance $\times$ Motion interaction, $\chi^2(4) = 12.89$, $p = .012$. Under jittering motion, saccades were faster toward Near ($p = .0016$) and Mid ($p = .0155$) than Far distractors, following a linear trend ($p = .0006$) (Fig. 4). Fixation duration showed a complementary interaction, $\chi^2(4) = 14.10$, $p = .007$, with longer fixations on Near ($p = .0001$) and Mid ($p = .0002$) under jitter.
- Distance perception was biased, $\chi^2(2) = 6.07$, $p = .048$: Near and Mid distances were underestimated ($p = .0026$; $p = .0499$), while Far was unbiased ($p = .503$). A main effect of response modality emerged, $\chi^2(1) = 12.23$, $p < .001$, with strong verbal underestimation ($\approx -52\%$) and accurate motor judgments.

4 DISCUSSION

Across two experiments, we show that action space is organized following a graded structure in VR. Experiment 1 showed that distractors presented in near space, and partially at intermediate distance, induced stronger behavioral interference, higher fixation probability, and faster saccadic orienting, revealing a spatial gradient centered on the region closest to the body. Experiment 2 confirmed

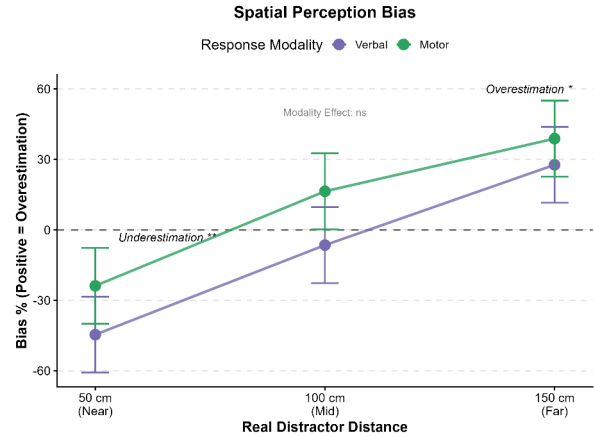


Figure 3: Exp.1 - Spatial perception bias across distances and response modalities. Near space is underestimated, while far space is overestimated.

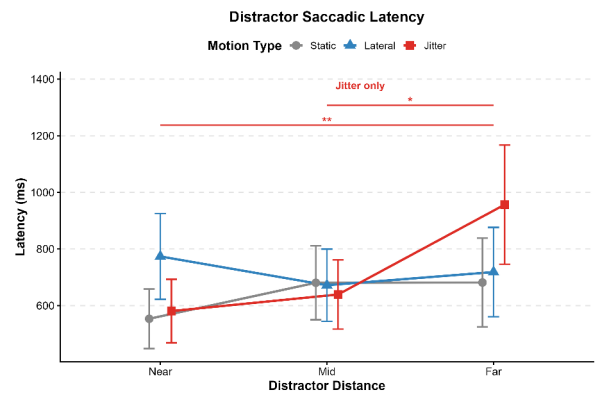


Figure 4: Exp.2 - Saccadic latency to the distractor as a function of distance and motion. Under jittering motion, saccades toward Near and Mid distractors are faster than those toward Far distractors.

this pattern, demonstrating that jittering motion selectively amplified distance-dependent modulations of saccadic latency and fixation duration. Distance judgments showed that distractors close to the body were systematically perceived as closer than their actual physical distance, whereas farther locations showed more variable and less biased estimates. Together, these findings show that action space is organized as a graded region rather than a fixed boundary. This baseline enables future work in Human Augmentation by testing whether extending the body's action capabilities shifts how this space is functionally represented.

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